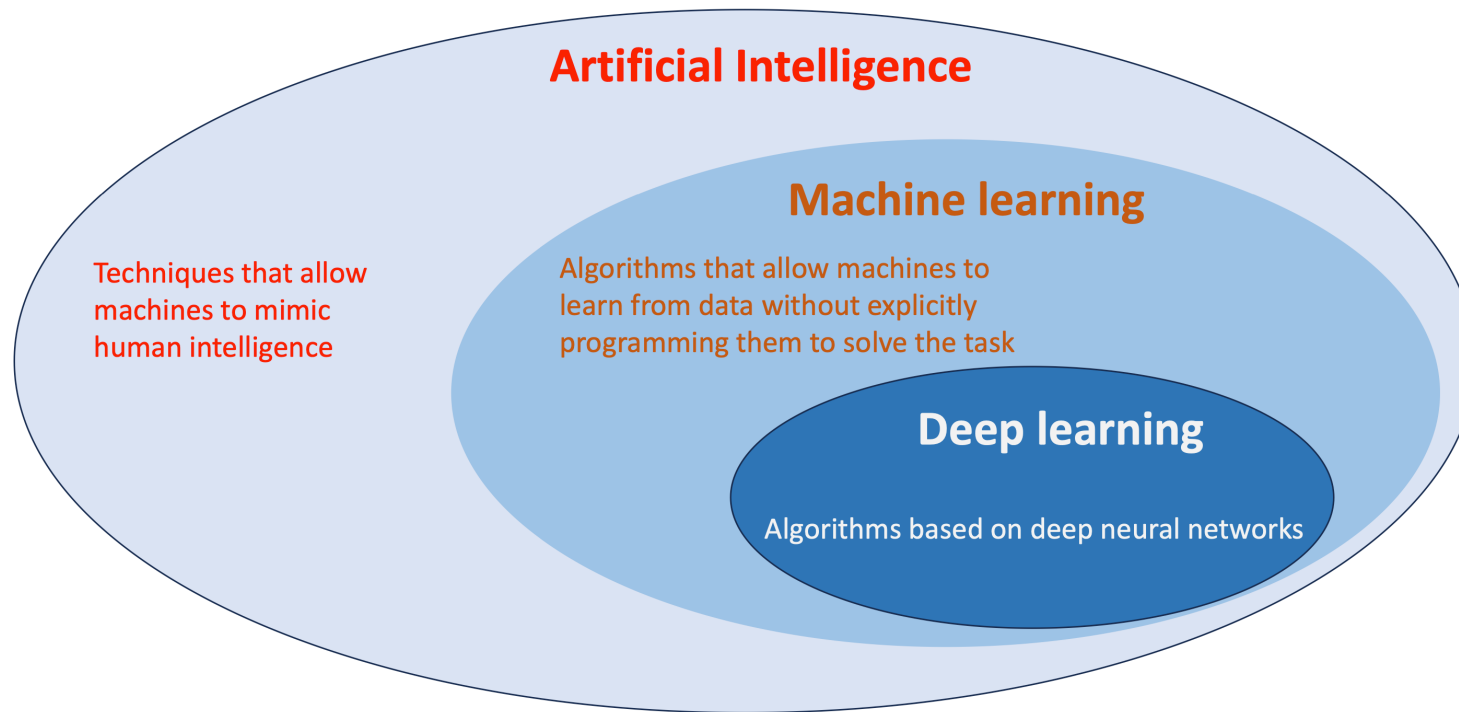

Deep Learning

Artificial Intelligence, Machine Learning, Deep Learning



Machine Learning: Regression

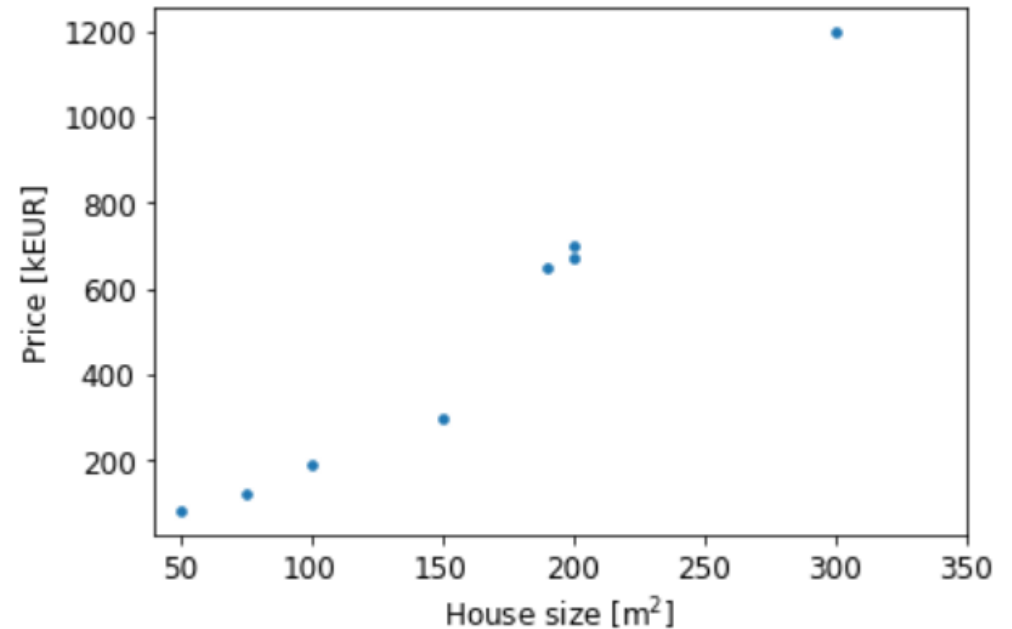
Real estate application

$\{(x^i, y^i)\} \quad i = 1, \dots, N$ available dataset

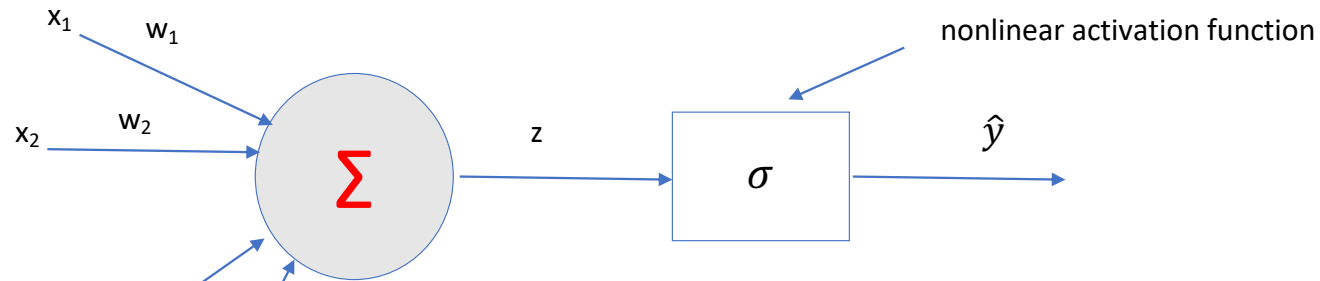
$\hat{y}^i = M(x^i; \theta)$ parametric model

$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N (y^i - \hat{y}^i(\theta))^2$ Loss (MSE)

$$\theta^* = \underset{\theta}{\operatorname{arg\,min}} \mathcal{L}(\theta)$$



Basic units for Neural Networks

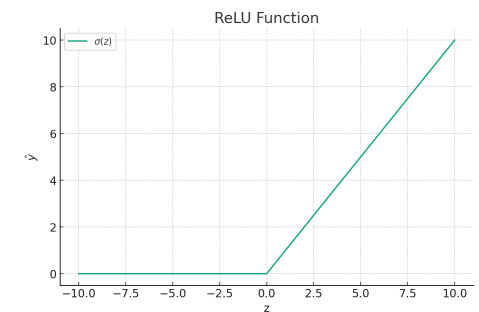
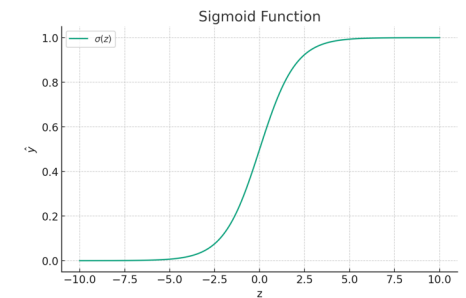


$$\hat{y} = \sigma \left(\sum_{j=1}^n w_j x_j + b \right)$$

$$\sigma(z) = \frac{1}{1 + e^{-z}} = \frac{e^z}{1 + e^z}$$

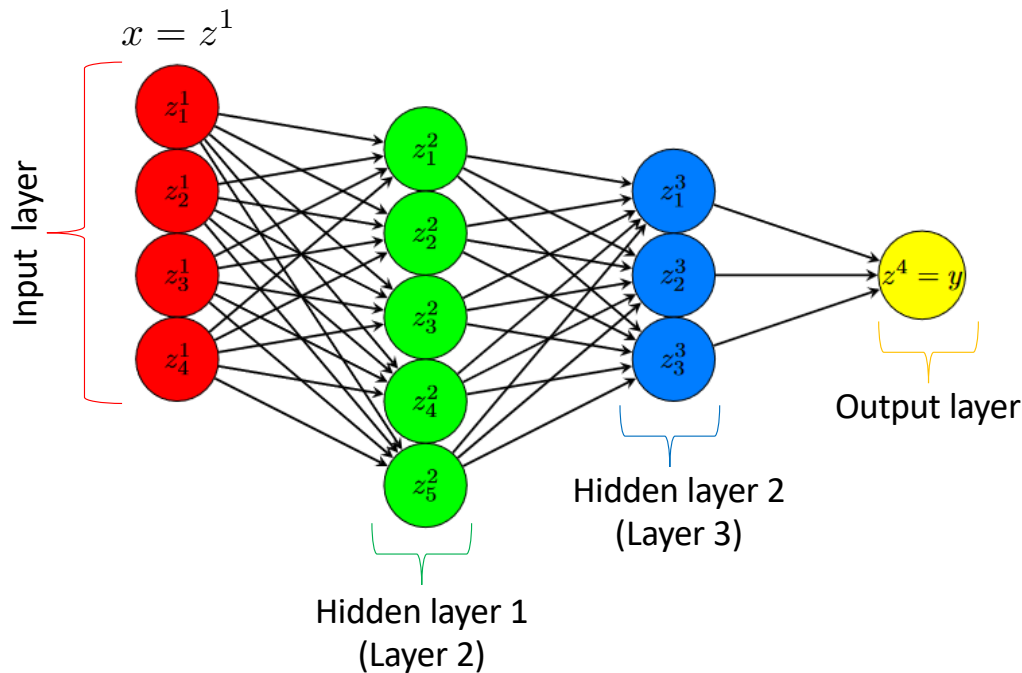
$$\sigma(z) = \begin{cases} z & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$$

We have a non-linear relation between inputs and outputs!



Fully-connected Feedforward Neural Networks

Move from one neuron to a hierarchical structure with fully connected neurons



$$z_1^2 = \sigma \left(\sum_{j=1}^4 w_{1,j}^1 z_j^1 + b_1^1 \right) = \sigma (W_1^1 z^1 + b_1^1)$$

$$z_2^3 = \sigma \left(\sum_{j=1}^5 w_{2,j}^2 z_j^2 + b_2^2 \right) = \sigma (W_2^2 z^2 + b_2^2)$$

$$y = z^4 = \sum_{j=1}^3 w_{1,j}^3 z_j^3 + b^3 = W_1^3 z^3 + b^3$$

$$\text{Overall: } y = W_3 \sigma (W_2 \sigma (W_1 x + b_1) + b_2) + b_3$$

We can easily define a NARX structures parameterized by the weights (and biases) of the network

$$\hat{y}(k) = f \left(\underbrace{y(k-1), \dots, y(k-na), u(k), u(k-1), \dots, u(k-nb)}_{z^1(k)}; W, b \right)$$

FFN: PyTorch

Definition of the model class

```
1 import torch
2 import torch.nn as nn
3
4 class FeedforwardNeuralNetModel(nn.Module):
5     def __init__(self, input_dim, hidden_dim, output_dim):
6         super(FeedforwardNeuralNetModel, self).__init__()
7
8         # Linear layer 1
9         self.fc1 = nn.Linear(input_dim, hidden_dim[0])
10        # Activation 1
11        self.sigmoid1 = nn.Sigmoid()
12
13        # Linear layer 2
14        self.fc2 = nn.Linear(hidden_dim[0], hidden_dim[1])
15        # Activation 2
16        self.sigmoid2 = nn.Sigmoid()
17
18        # Output layer (Linear layer)
19        self.output = nn.Linear(hidden_dim[1], output_dim)
20
21    def forward(self, x):
22        # Linear function # LINEAR
23        x = self.fc1(x)
24        x = self.sigmoid1(x)
25        x = self.fc2(x)
26        x = self.sigmoid2(x)
27        x = self.output(x)
28
29    return x
```

Instantiate the model class, run and show model

```
1 input_dim = 4
2 hidden_dim = [200, 300]
3 output_dim = 1
4 batch_dim = 10
5
6 model = FeedforwardNeuralNetModel(input_dim, hidden_dim, output_dim)
7
8 x = torch.randn((batch_dim, input_dim))
9 y = model(x)
10
11 from torchsummary import summary
12 summary(model, input_size=(input_dim,))
```

```
-----
          Layer (type)                Output Shape          Param #
=====
          Linear-1                    [-1, 200]              1,000
          Sigmoid-2                   [-1, 200]              0
          Linear-3                    [-1, 300]             60,300
          Sigmoid-4                   [-1, 300]              0
          Linear-5                    [-1, 1]                301
=====
Total params: 61,601
Trainable params: 61,601
Non-trainable params: 0
-----
Input size (MB): 0.00
Forward/backward pass size (MB): 0.01
Params size (MB): 0.23
Estimated Total Size (MB): 0.24
```

FFN: Training in PyTorch

```
# Define the model
model = FeedforwardNeuralNetwork(input_size=4, hidden_sizes=[200, 300], output_size=1)

# Define the Optimizer
optimizer = optim.SGD(model.parameters(), lr=1e-4)

# Batch Gradient Descent
for epoch in range(max_epochs):

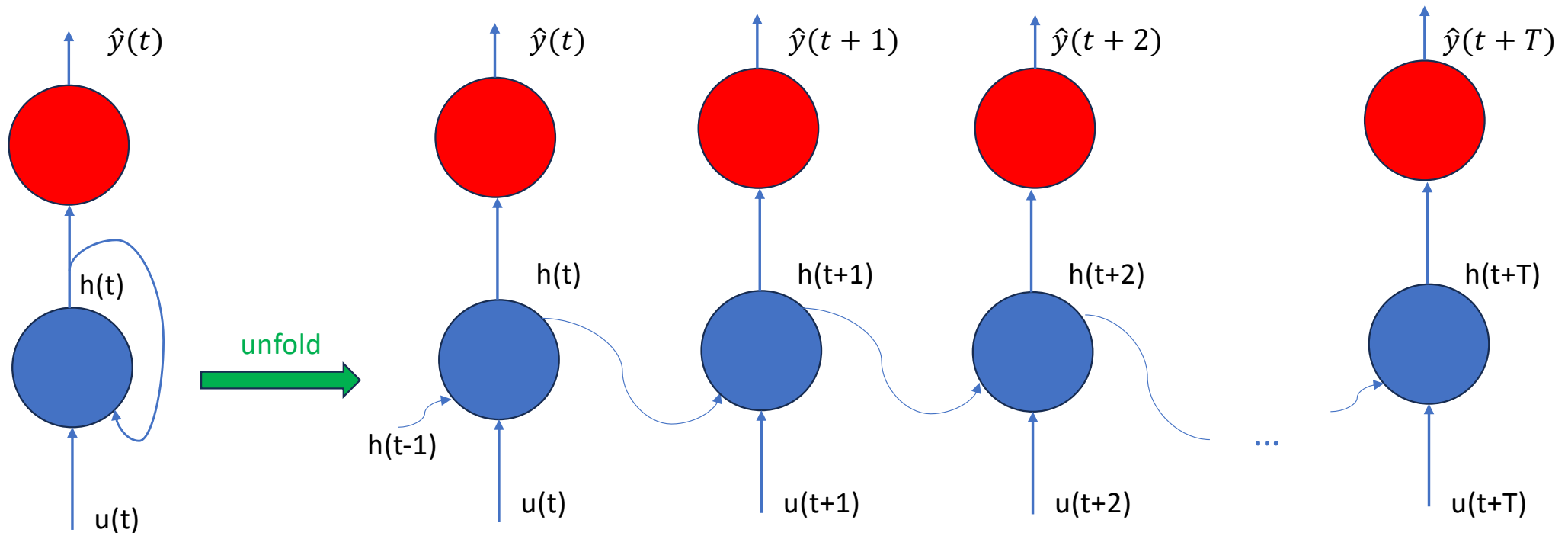
    optimizer.zero_grad()

    # forward pass
    y_hat = model(x)
    loss = torch.mean( (y_hat - y)**2 )

    # Backward pass and update
    loss.backward()
    optimizer.step()
```

Recurrent Neural Networks (RNNs)

- Architectures tailored to process time series data and temporal information (audio, text, video, signals, etc.)



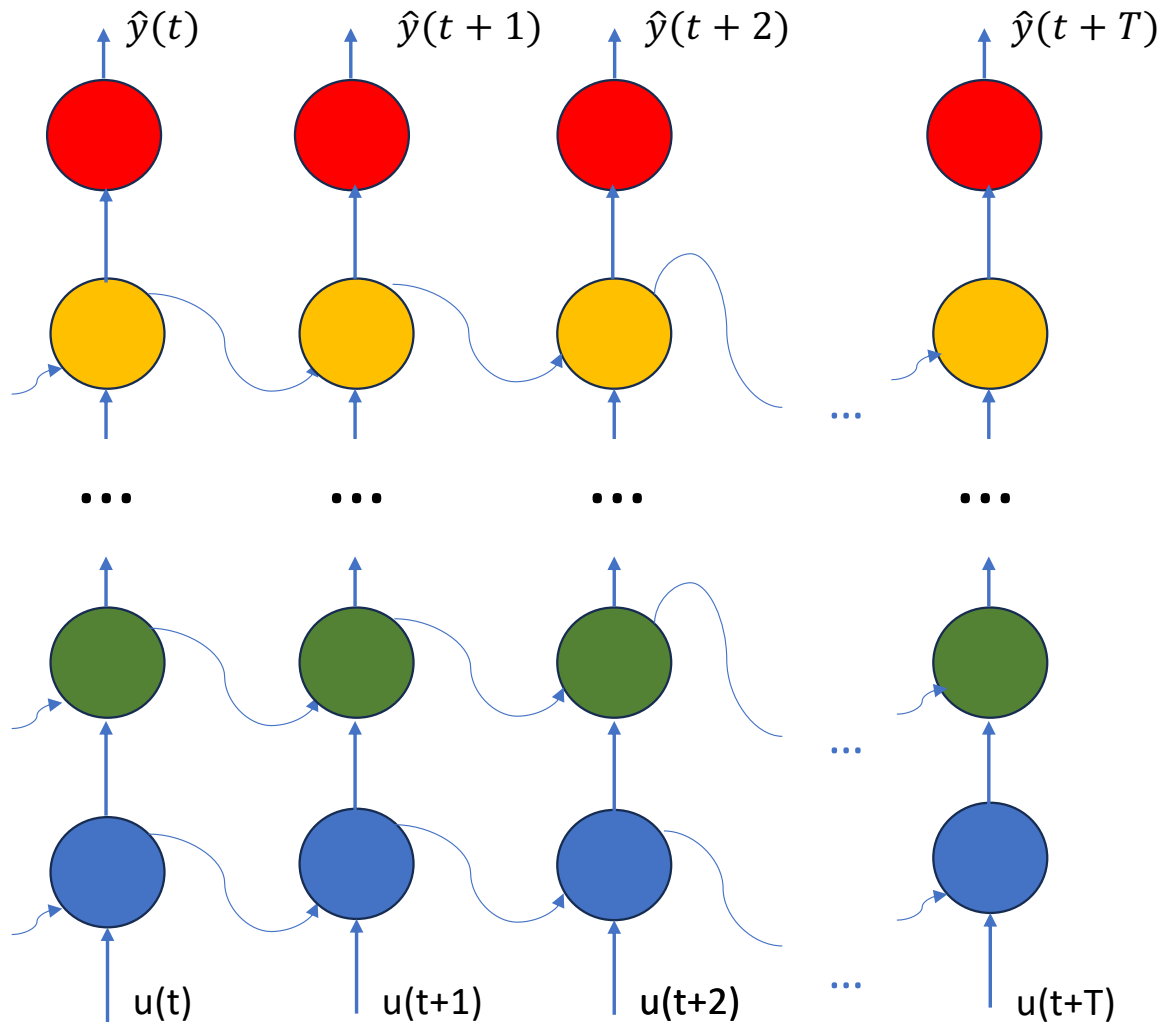
$$h(t) = f(h(t-1), u(t); W_f)$$

$$\hat{y}(t) = g(h(t); W_g)$$

$$\mathcal{L} = \sum_{i=0}^T \|\hat{y}(t+i) - y(t+i)\|^2$$

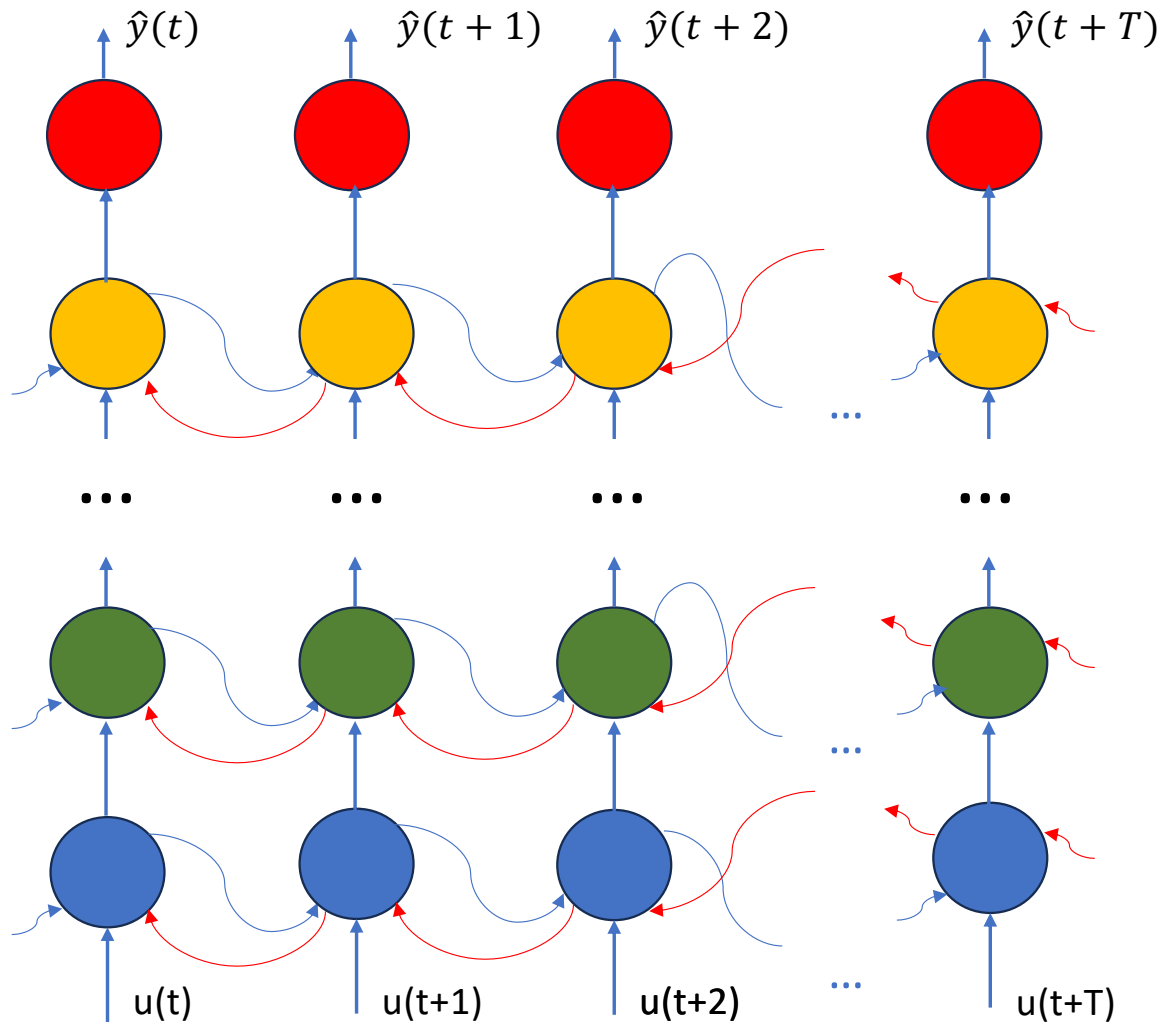
or skip some initial samples

Multi-layer RNN



Hidden states of a layer are also inputs of the next layer

Bidirectional RNN



Causality is lost. Might not be useful for prediction, but for tasks like smoothing

Creating training data

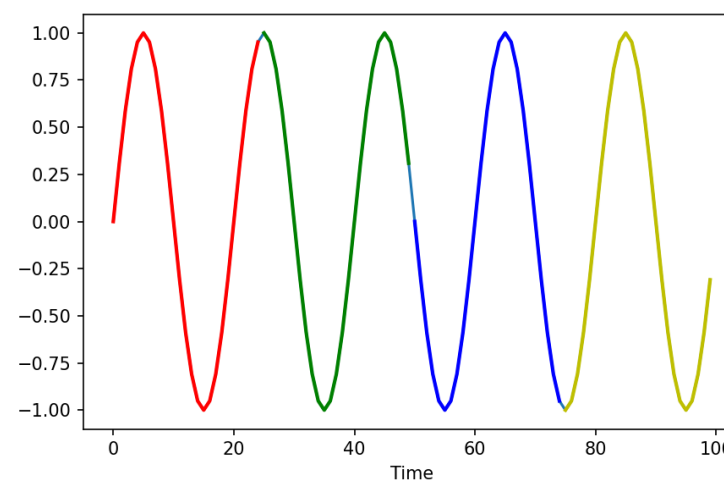
- Given an input/output sequence of length T , we can **unfold the network up to T steps** (you simulate T -step time ahead)

$$\mathcal{L} = \frac{1}{T} \sum_{t=0}^T \|\hat{y}(t) - y(t)\|^2$$

- ... or we can **split the sequence into shorter sub-sequences** (overlapped or not) of length $L \ll T$, and thus create batch of sub-sequences. **Network is unfolded for L steps** (when you train, you predict L -step time ahead)

$$\mathcal{L}^{(j)} = \frac{1}{L} \sum_{i=0}^L \|\hat{y}(t_j + i) - y(t_j + i)\|^2$$

$$\mathcal{L} = \frac{1}{B} \sum_{j=1}^B \mathcal{L}^{(j)}$$



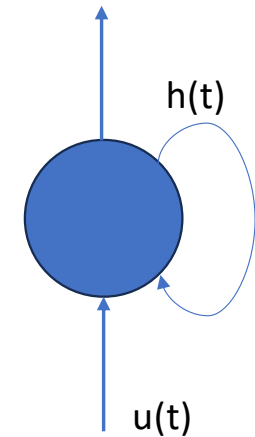
Vanilla Recurrent Neural Network

$$h(t) = \tanh(W_{hh}h(t-1) + W_{uh}u(t) + b_h)$$

```
CLASS torch.nn.RNN(self, input_size, hidden_size, num_layers=1,  
    nonlinearity='tanh', bias=True, batch_first=False, dropout=0.0,  
    bidirectional=False, device=None, dtype=None) [SOURCE]
```

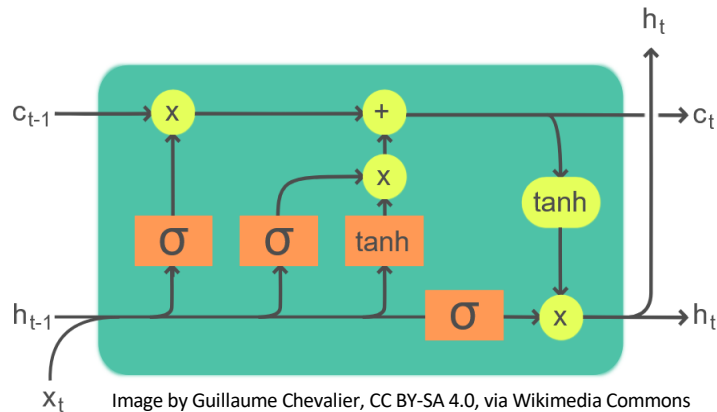
Parameters

- **input_size** – The number of expected features in the input x
- **hidden_size** – The number of features in the hidden state h
- **num_layers** – Number of recurrent layers. E.g., setting `num_layers=2` would mean stacking two RNNs together to form a *stacked RNN*, with the second RNN taking in outputs of the first RNN and computing the final results. Default: 1
- **nonlinearity** – The non-linearity to use. Can be either `'tanh'` or `'relu'`. Default: `'tanh'`
- **bias** – If `False`, then the layer does not use bias weights b_{ih} and b_{hh} . Default: `True`
- **batch_first** – If `True`, then the input and output tensors are provided as $(batch, seq, feature)$ instead of $(seq, batch, feature)$. Note that this does not apply to hidden or cell states. See the Inputs/Outputs sections below for details. Default: `False`



$$h(t) = f(h(t-1), u(t); W_f)$$

LSTM: Long Short-Term Memory Neural Networks

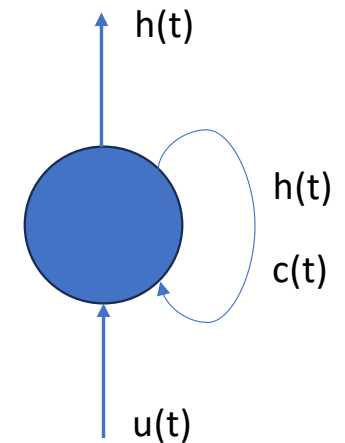


```
CLASS torch.nn.LSTM(self, input_size, hidden_size, num_layers=1, bias=True,
    batch_first=False, dropout=0.0, bidirectional=False, proj_size=0,
    device=None, dtype=None) [SOURCE]
```

```
B, L = 3, 5
u_size, hc_size, y_size = 10, 20, 1

lstm = nn.LSTM(u_size, hc_size, batch_first=True)
u = torch.randn(B, L, u_size)
h0 = torch.zeros(1, B, hc_size)
c0 = torch.zeros(1, B, hc_size)
h, (hn, cn) = lstm(u, (h0, c0))
L1 = nn.Linear(in_features = hc_size, out_features = y_size)
y = L1(hn)
print(h.shape, y.shape)

torch.Size([3, 5, 20]) torch.Size([3, 5, 1])
```



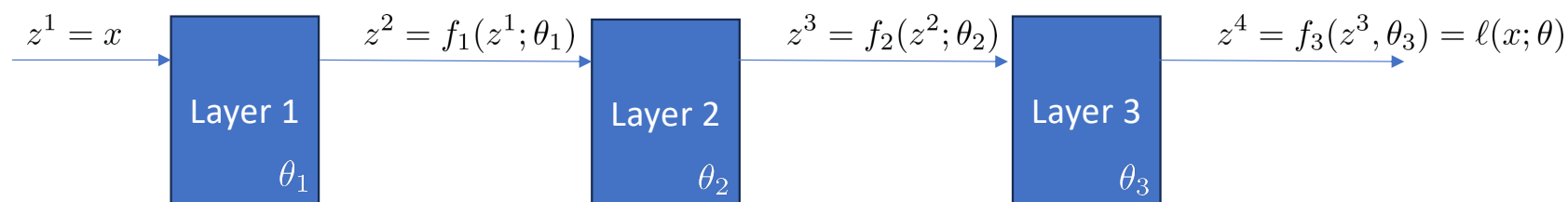
$$c(t) = f_c(c(t-1), h(t-1), u(t); W_c)$$

$$h(t) = f_h(c(t), h(t-1), u(t); W_h)$$

Backpropagation

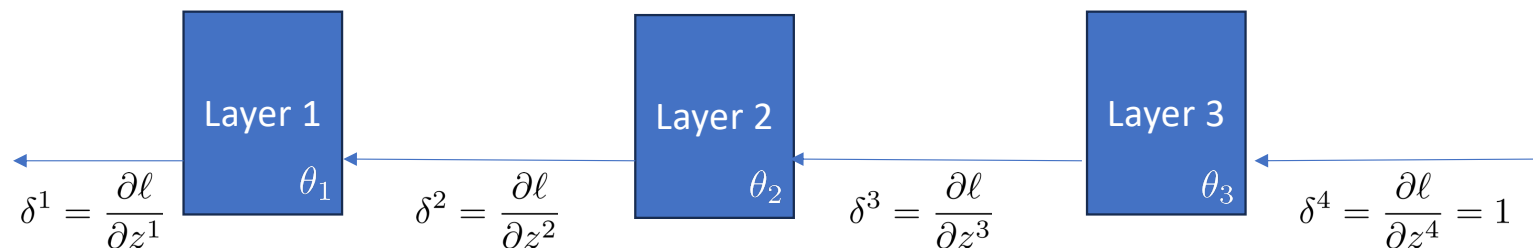
Back-propagation I

$$\ell(x; \theta) = (y_k - \hat{y}_k(x_k; \theta))^2$$



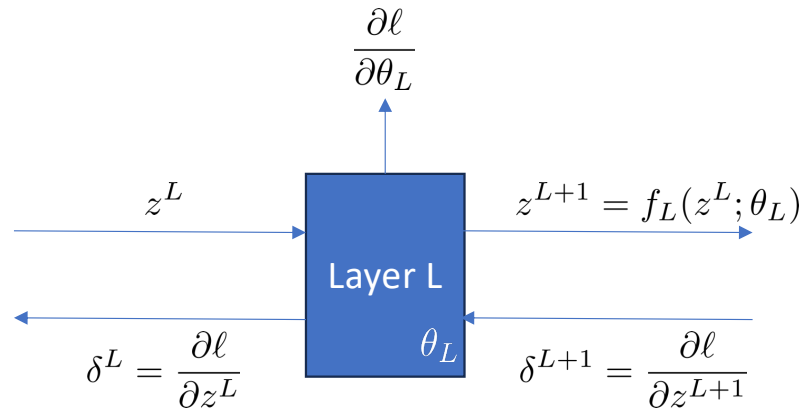
What we want: $\frac{\partial \ell}{\partial \theta_1}, \frac{\partial \ell}{\partial \theta_2}, \frac{\partial \ell}{\partial \theta_3}$

Idea: Back propagate $\delta^L = \frac{\partial \ell}{\partial z^L}$, $L = 4, 3, 2, 1$



Back-propagation II

Focus on Layer L



$$\delta^L = \frac{\partial \ell}{\partial z^{L+1}} \frac{\partial z^{L+1}}{\partial z^L} = \delta^{L+1} \frac{\partial z^{L+1}}{\partial z^L}$$

$$\frac{\partial \ell}{\partial \theta_L} = \frac{\partial \ell}{\partial z^{L+1}} \frac{\partial z^{L+1}}{\partial \theta_L} = \delta^{L+1} \frac{\partial z^{L+1}}{\partial \theta_L}$$

We have an interface to connect blocks/layers and recursively compute the derivatives!

Technicalities

Sigmoid activation function: $\sigma(z) = \frac{1}{1+e^{-z}} = \frac{e^z}{1+e^z}$; $\sigma'(z) = \sigma(z)(1-\sigma(z))$

Linear layer: $z^{L+1} = Wz^L$ $\frac{\partial z^{L+1}}{\partial z_i^L} = W_{:,i} \rightarrow \delta^L = \delta^{L+1}W$

```
for epoch in range(max_epochs):  
  
    optimizer.zero_grad()  
  
    # forward pass  
    y_hat = model(x)  
    loss = torch.mean( (y_hat - y)**2 )  
  
    # Backward pass and update  
    loss.backward()  
    optimizer.step()
```

Algorithm: reverse-mode automatic differentiation. Application to neural network training is called **back-propagation**.